

Talking Knots

All known written accounts of the Inka were set down after the conquest, most by Spaniards who had, of course, never experienced the empire in its heyday. Because many of the chroniclers tried to do their job conscientiously, most scholars use their reports, despite their deficiencies, as I do in this book. For obvious reasons historians of the Inka have never liked being forced to rely exclusively on postconquest, non-native written sources, but there seemed to be no avoiding it.

Recently, though, some researchers have come to believe that the Inka did have a written language—indeed, that Inka texts are displayed in museums around the world, but that they have generally not been recognized as such. Here I am referring to the bunches of knotted strings known as *quipu* (or *quipu*, as the term is often spelled). Among the most fascinating artifacts of Tawantinsuyu, they consist of a primary cord, usually a third to a half an inch in diameter, from which dangle thinner “pendant” strings—typically more than a hundred, but on occasion as many as 1,500. The pendant strings, which sometimes have subsidiary strings attached, bear clusters of knots, each tied in one of three ways. The result, in the dry summary of George Gheverghese Joseph, a University of Manchester mathematics historian, “resembles a mop that has seen better days.”

According to colonial accounts, *quipukamayuc*—“knot keepers,” in Ruma Suni—parsed the knots both by inspecting them visually and by running their fingers along them, Braille-style, sometimes accompanying this by manipulating black and white stones. For example, to assemble a history of the Inka empire the Spanish governor Cristóbal Vaca de Castro summoned *quipukamayuc* to “read” the strings in 1542. Spanish scribes recorded their testimony but did not preserve the *quipu*; indeed, they may have destroyed them. Later the Spanish became so infuriated when *quipu* records contradicted their version of events that in 1583 they ordered that all the knotted strings in Peru be burned as idolatrous objects. Only about six hundred escaped the flames.

All known writing systems employ instruments to paint or inscribe on flat surfaces. *Khipu*, by contrast, are three-dimensional arrays of knots. Although Spanish chronicles repeatedly describe *khipukamayuy* consulting their *khipu*, most researchers could not imagine that such strange-looking devices could actually be written records. Instead they speculated that *khipu* must be mnemonic devices—personalized memorization aids, like rosaries—or, at most, textile abacuses. The latter view gained support in 1923, when science historian L. Leland Locke proved that the pattern of knots in most *khipu* recorded the results of numerical calculations—the knotted strings were accounting devices. *Khipu* were hierarchical, decimal arrays, Locke said, with the knots used to record is on the lowest level of each string, those for the tens on the next, and so on. “The mystery has been dispelled,” archaeologist Charles W. Mead exulted, “and we now know the quipu for just what it was in prehistoric times . . . simply an instrument for recording numbers.”

Based on such evaluations, most Andeanists viewed the Inka as the only major civilization ever to come into existence without a written language. “The Inka had no writing,” Brian Fagan, an archaeologist at the University of California in Santa Barbara, wrote in *Kingdoms of Gold, Kingdoms of Jade*, his 1991 survey of Native American cultures. “The quipu was purely a way of storing precise information, a pre-Columbian computer memory, if you will.”

But even as Fagan was writing, researchers were coming to doubt this conclusion. The problem was that Locke’s rules only decoded about 80 percent of *khipu*—the remainder were incomprehensible. According to Cornell archaeologist Robert Ascher, those *khipu* are “clearly non-numerical.” In 1981, Ascher and his mathematician wife, Marcia, published a book that jolted the field by intimating that these “anomalous” *khipu* may have been an early form of writing—one that Ascher told me was “rapidly developing into something extremely interesting” just at the time when Inka culture was demolished.

The Aschers slowly gained converts. “Most serious scholars of *khipu* today believe that they were more than mnemonic devices, and probably much more,” Galen Brokaw, an expert in ancient Andean texts at the State University of New York in Buffalo, said to me. This view of *khipu* can seem absurd, Brokaw admitted, because the scientists who propose that Tawantinsuyu was a literate empire also freely admit that no one can read its documents. “Not a single narrative *khipu* has been convincingly deciphered,” the Harvard anthropologist Gary Urton conceded, a situation he described as “more than frustrating.”

Spurred in part by recent insights from textile scholars, Urton has been mounting the most sustained, intensive attack on the *khipu* code ever performed. In *Signs of the Inka Khipu* (2003), Urton for the first time systematically broke down *khipu* into their grammatical constituents, and began using this

catalog to create a relational *khipu* database to help identify patterns in arrangement of knots. Like cuneiform marks, Urton told me, *khipu* probably did begin as the kind of accounting tools envisioned by Locke. But by the time Pizarro arrived they had evolved into a kind of three-dimensional binary code unlike any other form of writing on earth.

The Aschers worked mainly with *khipu* knots. But at a 1997 conference William J. Conklin, a researcher at the Textile Museum, in Washington, D.C., pointed out that the knots might be just one part of the *khipu* system. In interview, Conklin, perhaps the first textile specialist to investigate *khipu* explained, “When I started looking at *khipu* . . . I saw this complex spinning and plying and color coding, in which every thread was made in a complex way. I realized that 90 percent of the information was put into the string before the knot was made.”

Building on this insight, Urton argued that *khipu* makers were forced the very nature of spinning and weaving into making a series of binary choices, including the type of material (cotton or wool), the spin and direction of the string (which he described as “S” or “Z,” after the “slant” the threads), the direction (recto or verso) of the knot attaching the pendant string to the primary, and the direction of the main axis of each knot itself (or Z). As a result, each knot is what he called a “seven-bit binary array, although the term is inexact because *khipu* had at least twenty-four possible string colors. Each array encoded one of $2^6 \times 24$ possible “distinct informational units”—a total of 1,536, somewhat more than the estimated 1,000 to 1,500 Sumerian cuneiform signs, and more than twice the approximately 600 to 800 Egyptian and Maya hieroglyphic symbols.

If Urton is right, *khipu* were unique. They were the world’s sole intrinsically three-dimensional written documents (Braille is a translation of writing on paper) and the only ones to use a “system of coding information” that “fit the coding systems used in present-day computer language, was structured primarily as a binary code.” In addition, they may have been among the few examples of “semasiographic” writing—texts that, unlike written English, Chinese, and Maya, are not representations of spoken language. “A system of symbols does not have to replicate speech to communicate narrative,” Catherine Julien, a historian of Andean cultures at Western Michigan University, explained to me. “What will eventually be found in *khipu* is uncertain, but the idea that they have to be a representation of speech has to be thrown out.”

Writing and reading are among the most basic methods of transmitting information from one person to another. In cultures throughout the world this procedure is fundamentally similar. One reads a parade of symbols, taking up information with the eyes; emphasis and context is provided visually by changing the size and form of the symbols (printing in italics or boldface

increasing or diminishing the font size, scattering words or characters around the page). All European and Asian cultures share the common experience of reading—sitting in a chair, the book in one's lap, wagging the head from side to side (Europe) or up and down (Asia).

Because Tawantinsuyu existed only for a few centuries, it is widely assumed that the Inka *kipu* built on other, earlier forms of writing that had been developed in the region. And these cultures were unique, if Urton is right. Their books were loose bundles of string—more practical, in some ways, than paper scrolls or books, because less susceptible to water damage and physical pressure. They were read both tactilely, by running the fingertips along the knots, and visually, by looking at the colors of the strings. And whereas the choice of letters and words at the beginning of a sentence or paragraph exercise little constraint on physical connection to those at the end, the choices made by the *kipu* maker at the beginning of a string could not be undone halfway through. As a result, each *kipu* pendant provided a burst of information at the beginning that was refined further down the string.

However anomalous to European eyes, this form of writing has deep roots in Andean culture. Knotted-string communication was but one aspect of these societies' exploration of textile technology (see Chapter 3). In these cultures, Heather Lechtman, of MIT, has argued, cloth "was the most important carrier of status, the material of choice for the communication of message, whether religious, political, or scientific." Similarly, Urton told me, binary oppositions were a hallmark of the region's peoples, who lived in societies "typified to an extraordinary degree by dual organization," from the division of town populations into complementary "upper" and "lower" halves (moieties, in the jargon) to the arrangement of poetry into dyadic units. In this environment, he said, "*kipu* would be familiar."

As this book went to press, Urton and Carrie Brazine, a software developer/weaver/mathematician at Harvard's Peabody Museum, were creating an online database characterizing most of the surviving *kipu*, a more sophisticated successor to a pioneering database created by Robert and Marcia Archer. The hope is that computerized analysis will be able to spot patterns that thus far have eluded human researchers.

At the same time, Urton and other *kipu* specialists have been searching for an Inka Rosetta stone—a colonial translation of an extant *kipu*. One candidate exists—maybe. In 1996, Clara Miccinelli, an amateur historian from the Neapolitan nobility, caused a stir by announcing that she had unearthed in her family archives both a *kipu* and its Spanish translation (it encoded a folk song). But because the putative *kipu* isn't made the same way as other surviving *kipus* and the same documents also claim that Pizarro conquered the Inka empire by poisoning its generals with arsenic-adulterated wine, many

U.S. scholars have questioned their authenticity. Angered by the doubts, Miccinelli has thus far refused to let non-Italian researchers examine the documents, although she did allow an Australian laboratory to evaluate their age with a mass spectrometer. (The results, published in 2000, suggest that they are from the fifteenth century.) Because of the controversy, most researchers have been, according to Brokaw, "strategically ignoring" the Italian documents, at least for the present.

More widely accepted are the thirty-two *kipu* found in a tomb in the Peruvian Amazon in 1996, one of which Urton tentatively deciphered as a census record for the area in late pre-Hispanic times. With the help of a MacArthur fellowship he received in 2001, he has been searching Peruvian archives for something with more narrative content to match against the other *kipu*—a quest, according to Julien, that "has a chance of bearing fruit." If Urton's quest or others like it are successful, she told me, "We may be able to hear the Inkas for the first time in their own voice."

I asked what she thought that voice might sound like—the voice of people attuned to tension and cloth, people who saw the stones of the world charged with spirit, people who had never seen animals larger than a llama, people who broke the world into complementary halves and thought more in terms of up and down than north and south, people who took in information about the world through their fingers.

"Foreign," she said.